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The Magnetization of Lower Carboniferous Rocks from the North-west Border of the Dartmoor Granite, Devonshire

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Summary

The directions of magnetization and magnetic properties of lower Carboniferous rocks from the north-western border of the Dartmoor granite are as complex as the geological history of the area. In the Okehampton district and particularly in the Meldon Quarries it was found that the directions of magnetization are grouped in a way which cannot be explained by either the folding or by reheating *in situ*. The directions are also anomalous when compared with the results obtained elsewhere for the Carboniferous and early Permian.

Evidence from thermal and a.c. demagnetization experiments, thermomagnetic balance results and the mineralogy suggests that the remanent magnetization of many of the igneous and sedimentary rocks is due to pyrrhotite, formed during the intrusion of the Dartmoor granite. The pyrrhotite gives rise to an anisotropy of susceptibility which is probably responsible for some of the anomalous directions of magnetization, but it is conceivable that post-granite fault movements also caused some remagnetization.

Pyrrhotite is present in lesser amounts in rocks of the Tavistock district. The mean directions for four sites have been found to be similar, relative to the present horizontal, despite great differences in the field positions of the rocks. Their total mean ($189° - 21°$) is similar to the result obtained from the Exeter lavas, Devonshire, which are identical in age to the Dartmoor granite. It is believed that the remagnetization was the combined result of heating due to the intrusion of the granite and the simultaneous introduction of pyrrhotite by metasomatic action.

1. Introduction

The magnetization of the Lower Culm (Carboniferous) sediments and igneous rocks of the Tavistock and Okehampton districts, Devonshire, is as complex as the geological history of these highly disturbed rocks lying around the north-western margin of the Dartmoor granite. The original intention of this survey was to discover, by palaeomagnetic methods, the age of the dolerite intrusions relative to the period of folding and to decide if the apparent displacement of the Culm Measures was a result of the intrusion of the Dartmoor granite. For the first problem the conclusion reached depends upon whether the directions of magnetization show lower scatter before or after the dip correction has been made, while for the second problem any large horizontal displacement of the rocks would result in a rotation of the directions

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relative to those of contemporaneous rocks from stable areas. As the rocks were known to have been involved in contact metamorphism it was anticipated that thermal demagnetization would be necessary to discover the original directions of magnetization.

It was found, however, that neither of these problems could be solved as the remanent magnetization is due mainly to the pyrrhotite formed by metasomatism connected with the intrusion of the granite. The magnetic properties of these altered sediments and dolerites form the basis of this paper.

Although seventeen sites along the north-west border of the Dartmoor granite were visited, the most intensive sampling was made at two quarries—the Meldon Quarries in the north near Okehampton, and the Pitts Cleave Quarry, 11 miles to the south and two miles north of Tavistock. The magnetization of the rocks from the Okehampton and Tavistock districts and from these two quarries in particular differ although the rock types and geological environment are similar. The sites have therefore been divided geographically into two groups and the results will be described separately.

2. Geology

The regional settings of the Okehampton and Tavistock districts and the positions of the collecting sites are shown in Figs. 1 and 11. The main structural feature of this part of south-west England is the east–west syncline of Central Devon. The centre of this complex fold is occupied by Upper Culm Measures, flanked on both sides by narrow zones of the Lower Measures. The Culm is underlain by Devonian sediments which are of marine origin in the south but of mixed marine and continental origin in the north of the county.

The Okehampton district lies at a point where the regional strike of the Culm Measures has been apparently deflected by the intrusive mass of the Dartmoor granite. The strike deflection can be judged from the shape of the northern margin of the Lower Culm Measures shown in the small map in Fig. 1. In the Tavistock district there has

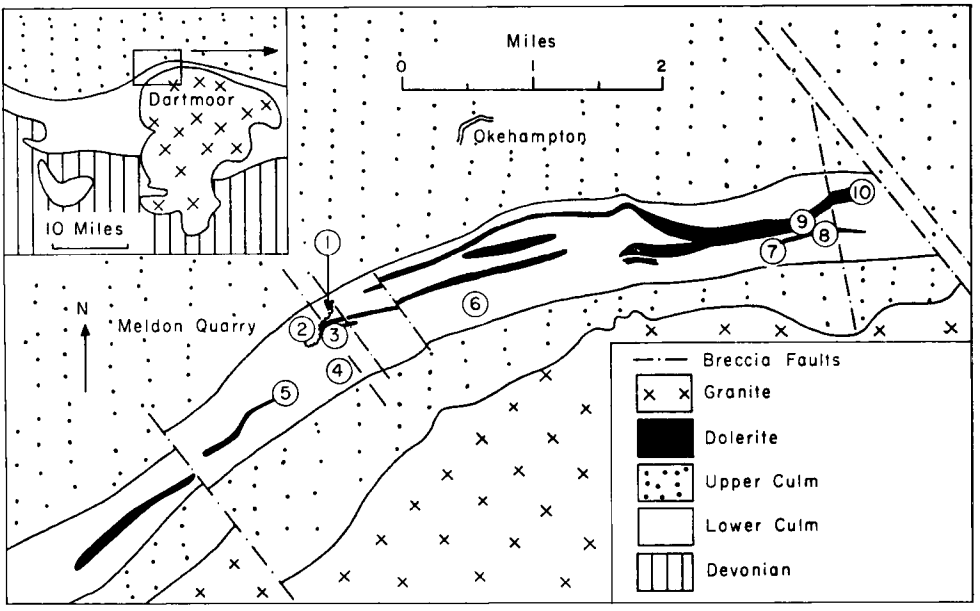


FIG. 1. Geological map of the Okehampton district, Devonshire, showing the positions of the collecting sites 1–10. (After Dearman & Butcher 1959.)

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been little horizontal displacement and the southern limb of the syncline is abruptly terminated by the granite. The stratigraphical succession and structure along the north-west border of the Dartmoor granite have been described by Dearman & Butcher (1959). Dearman (1959) has made a very detailed examination of the structures exposed at successive intervals during the expansion of the Meldon Quarries, Okehampton.

The oldest rocks recognized by Dearman (1959) in the Okehampton area belong to the Slate-with-lenticles Group and are of supposed Devonian age. The lower part of the Lower Culm is made up of the black shales with small lenticles of quartzite known as the Meldon Shale and Quartzite Group. This has been divided into lower and upper parts by the thick, but local, development of tuffs and porphyries. The upper part of the Lower Culm is characterized by a series of black, impure limestones with subordinate shales and cherts. The total thickness of the Lower Culm is about 700 ft. The Upper Culm Measures are at least 1300 ft thick in this area and consist of shales and grits. In the Tavistock district the succession is similar but the Lower Culm Measures occupy a greater proportion of the area (Fig. 11).

Both the Tavistock and the Okehampton districts contain minor intrusions of basic composition but there is a fundamental difference in their ages relative to that of the folding according to the descriptions of Dearman & Butcher. In the Okehampton district two sheets of albite–dolerite, referred to as the Type I and Type II intrusions by Dearman (1959), have been emplaced parallel to the bedding planes of the Lower Culm sediments. Dearman believes them to be younger than the main period of Hercynian orogenic deformation but older than the Dartmoor granite. Miller & Mohr (1964) subsequently showed the intrusions to be 296 and 295 m.y. old, compared with the age of 280 m.y. of the granite. Basic intrusions are far more common in the Tavistock district and there is also strong evidence to show that they are true sills and were fully involved in the fold movements. Dearman & Butcher (1959) even suggest that they were intruded while the sediments were still wet and unconsolidated. The sills are medium to fine grained oligoclase–dolerite varying in thickness between a few inches and 200–300 ft.

The structural pattern of the entire north-west border of the Dartmoor granite is determined by the presence of zig-zag folds formed during the main period of orogeny and later modified during the intrusion of the granite. In the Tavistock district the folds are recumbent but northwards they have been tilted at successively greater angles to the west or north by the granite until in the Okehampton district the fold axial planes dip away at 45° from the granite margin. In the latter district the main structural feature is an anticline of Lower Culm Measures which runs parallel to the granite margin but is separated from it by a narrow syncline of Upper Measures. As the folds are overturned to the south or south-east the general dip of the sediments is to the north or north-west, away from the granite. In the Meldon Quarries the Lower Culm anticline is divided into two minor folds by a central syncline which dies out to the east (Fig. 2). The structure of the Okehampton area is further complicated by the presence of bedding plane faults, jointing and breccia faults. The latter strike north-west to south-east (Fig. 1) and are the results of the final phase of deformation in the area. They coincided with the late hydrothermal stage of the Dartmoor granite and are sometimes important as the site of mineral veins. Perhaps of greater significance to the present study is the fact that these faults are slightly younger than the period of metasomatism.

The thermal aureole of the Dartmoor granite is about 1500 ft thick and includes many of the sites visited in this survey. This metamorphism was closely followed by a period of metasomatism resulting from the release of reactive emanations from the granite along joints and other structural planes. The products of this action are numerous, particularly in the Meldon Quarries, but the most important from the point of view of this study is pyrrhotite. This mineral occurs both in veins and as

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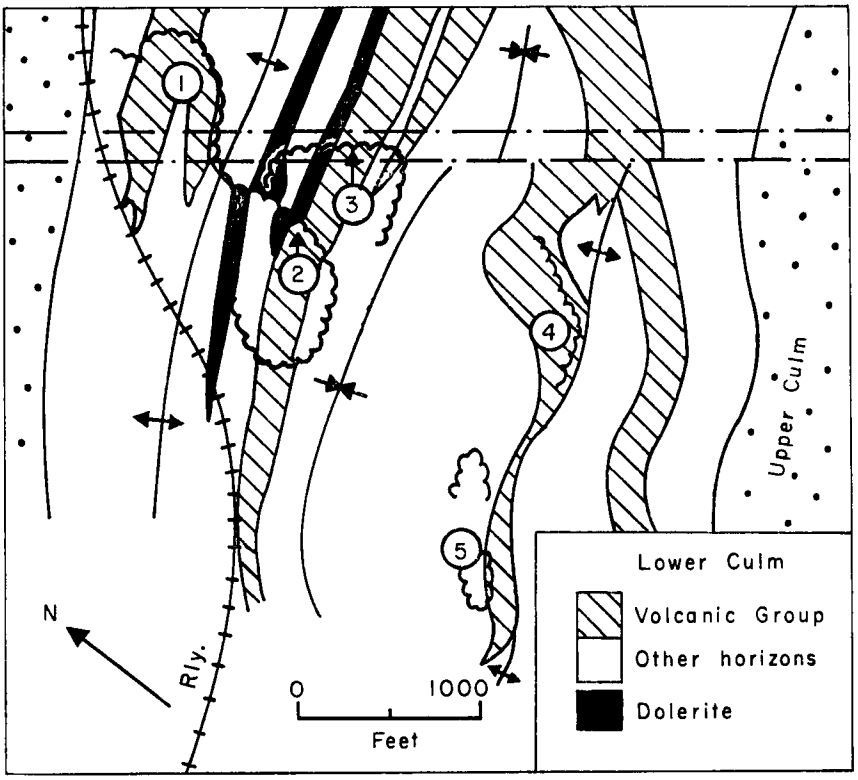


FIG. 2. Simplified geological map of the Meldon area showing the positions of collecting sites 1–5. (After Dearman 1959.)

impregnations in the country rock and appears to have been formed on several occasions during the metasomatism. Pyrrhotite was also one of the products of a localized final phase of iron metasomatism centred at the Red-a-Ven Brook, half a mile south of the main Meldon Quarries (Dearman & Butcher 1959). The last event in the history of the Dartmoor granite was the hydrothermal activity which gave rise to the mineral lodes and to the mineralization of the simultaneously formed breccia faults.

3. Measurements of the magnetization

Orientated hand samples collected at the sites listed in Table 1 were machined to provide an average of four specimens per sample. The directions and intensities of the remanent magnetization were measured with an astatic magnetometer and the results combined to give the sample means. These in turn were combined to yield the site means listed in Table 1. Because the intensities of both the igneous and the sedimentary rocks are often rather low many specimens were measured on the astatic magnetometer normally used for sedimentary rocks. The susceptibility of at least one specimen from each sample was measured in a field of 10 oersteds with a transformer bridge and these results have also been included in the table. The stability of the remanent magnetization was examined by the standard procedures of thermal and a.c. demagnetization and its origin was decided after the examination of thermomagnetic curves and polished sections.

A direction of magnetization has been plotted as an open circle if the north seeking pole lies above the horizontal and as a full dot if it lies below the horizontal on an

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equal area projection. The scatter of the directions (unit vectors) has been analysed according to the method of Fisher (1953), in which α is the half angle of the 95% cone of confidence and k is an estimate of the precision factor K.

4. The remanent magnetization of samples from the Okehampton district

4.1. *The NRM and susceptibility*

The Okehampton district lies to the north of the Dartmoor granite and the collecting sites 1–10 are all situated within the narrow anticline of Lower Culm Measures (Fig. 1). Within this area the most intensive sampling was carried out in the Meldon Quarries where the large excavations reveal the entire succession of Lower Culm Measures and the dolerite intrusions at a point where the overturned anticline is divided into two by a central syncline (Fig. 2). The various bays of the Meldon Quarries are referred to by the elevation of the working levels—the 860, 950 and 1010 (ft) levels being the most important (sites 1 and 2, 3 and 4, respectively, in Table 1). The investigations described on the following pages were performed mainly on samples from these quarries.

Data on the magnetization of the samples from the Meldon district have been summarized in Table 1. The site mean directions and circles of confidence for the sediments, the Type I intrusion and the Type II intrusion from the Meldon Quarries (sites 1–5) are shown in Fig. 3. No dip corrections for the sediments have been made in this figure, but it will be shown that such a correction is not necessary. It is obvious

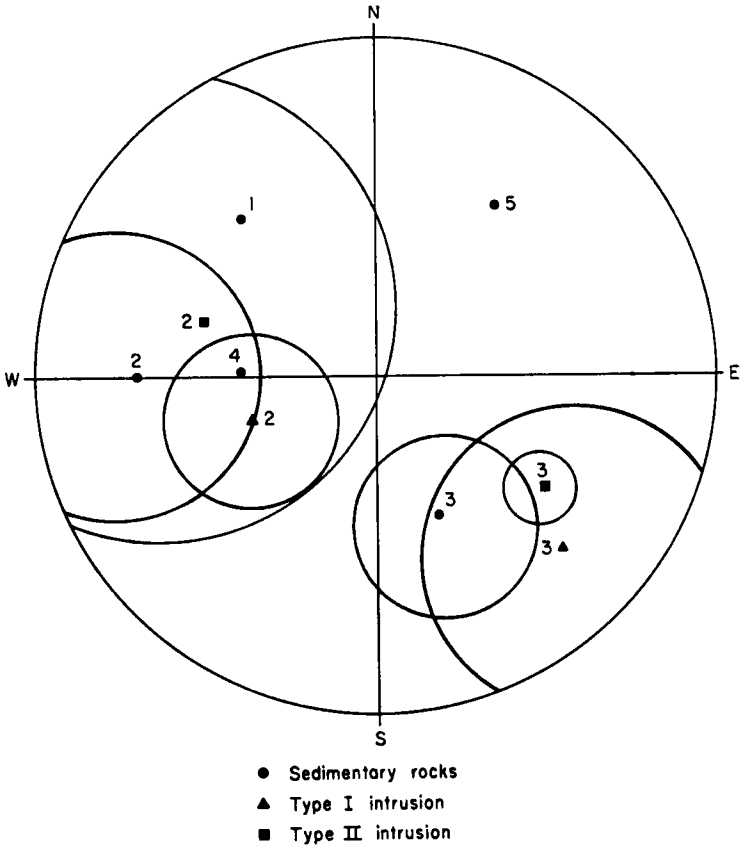


FIG. 3. Mean directions of magnetization of the sediments and the two basic intrusions at sites 1–5, Meldon. The circles of confidence of the means for sites 2 and 3 only are shown. Black symbols are used for directions on the lower hemisphere and open symbols (absent in this diagram) for directions on the upper hemisphere.

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Table 1

Directions of magnetization and properties of samples from the Okehampton and Tavistock districts

Site No.	Locality	National grid reference	Rock types	Mean directions $D°$	$I°$	Scatter $\alpha°$	k	Intensity $\times 10^{-6}$ gauss	Suscept. $\times 10^{-6}$ gauss/ oersted	Q	N
Okehampton district											
Meldon Quarries											
1	860 N bay	570929	Sediments	320	+29	—	—	174	50	7·7	2
2	860 S bay	569925	Sediments	270	+20	32	3	145	112	2·9	10
			Dolerite I	250	+49	25	16	927	478	4·1	5
			Dolerite II	288	+35	57	4	308	322	2·0	4
3	950 S bay	569925	Sediments	156	+41	25	4	171	30	12·5	12
			Dolerite I	132	+17	34	4	420	236	4·0	7
			Dolerite II	123	+29	9	38	11238	157	159·2	7
4	1010 bay	571921	Sediments	271	+47	46	3	211	187	2·3	7
5	Aplite Quarry	567920	Sediments	34	+28	42	4	4004	274	32·2	5
Okehampton											
6	Okehampton camp	591928	Sediments	Scattered		—	—	101	21	10·7	5
7	Belstone	625936	Sediments	Scattered		—	—	4	17	0·5	3
8	River Taw	630937	Dolerite I	159	+12	26	8	185	127	3·2	6
9	Sticklepath	631939	Dolerite I	Scattered		—	—	145	23	14·0	4
10	Sticklepath Quarry	637943	Dolerite I	210	−25	19	41	52	52	2·2	3
Tavistock district											
11	Smeardon Down	522779	Sediment	186	−19	—	—	348	178	4·4	1
12	Smeardon Down	518779	Dolerite	192	−21	31	17	6	59	0·2	4
13	Hill Bridge	532804	Dolerite	172	−16	30	18	7	55	0·3	3
14	Pitts Cleave	502762	Dolerite	198	−15	14	10	543	88	13·7	13
			Sediment	181	−28	16	23	17	13	2·9	5
15	Wilminstone	490755	Dolerite	207	+10	59	2	31	48	1·4	6
			Sediment	196	−34	18	47	2	26	0·2	3
16	Cox Tor	530760	Dolerite	Scattered				1771	52	76·0	3
17	Tavistock	489744	Lavas	Too weak				<1	45	—	3

from Fig. 3 that a broad division of the directions into two groups can be made on the basis of the declination of the means shown. The inclination is always downwards and has intermediate to steep values. The first group of directions is characterized by south-east declinations and will be referred to as Group A. These directions are shown by samples of sediments, Type I dolerite and Type II dolerite from the 950 S bay. The Group B directions have declinations of about 270° and are characteristic for samples of sediments and both igneous intrusions from the 860 S bay and for the sediments from the 1010 level bay. The directions of samples from the Aplite Quarry however do not correspond to either of these groups. The sediments from the 860 N bay were mostly too weakly magnetized to produce significant results but the two which were measured appear to belong to Group B.

This division of the directions of magnetization is not the result of the extreme degree of folding seen in the quarries. This can be shown most clearly by comparing the results from the 860 S and 950 S level bays. Samples of both sediments and igneous rocks were collected from the north-east face of the two bays at points only 90 ft apart vertically and about 300 ft apart in the horizontal plane. The two quarry bays in fact expose the same stratigraphical horizons at two levels in the overturned limb of a fold. The difference of about 90° in the Group A and B directions of the rocks from the 950 S and 860 S level bays cannot therefore be explained by small differences in the dip at the two sites. It is also obvious from the fact that there are two distinct groups of directions that remagnetization as a result of the metamorphism offers no complete explanation.

The accuracy of the mean directions in the Meldon quarries varies considerably, as can be seen from the circles of confidence in Fig. 3. There is a remarkably low scatter for the Type II dolerite sample directions from the 950 S bay while the same intrusion in the 860 S bay appears to have an unstable magnetization. At the latter site samples from the Type I intrusion display the lowest scatter. The sediments generally have quite variable directions and even samples collected from within the contact zones of the intrusions do not agree particularly well with the direction of the intrusion itself. The circles of confidence have been calculated using the data from all the samples measured and could in some cases be reduced in size by the omission of one or two anomalous values.

The samples from the Meldon quarries are also remarkable for the occasional very large intensity and susceptibility values (Fig. 4) and for the variations in these properties over short distances. Four adjacent specimens cut from one sample of the Type I intrusion, for example, were found to have intensities of 1262, 465, 209 and 1×10^{-6} gauss. Although the sediments generally have lower intensities and susceptibilities than the intrusions the values recorded (Table 1) are still remarkably high for these rock types. Intensities of less than 10×10^{-6} gauss and a mean susceptibility of 5×10^{-6} gauss/oersted were found from samples of cherts, shales and limestones similar to those at Meldon but collected at sites remote from the granite. A general explanation for these high and variable intensities as remagnetization due to lightning is unlikely because the sites are up to half a mile apart and are in quarries which are being worked at the present time.

The Koenigsberger ratio (Q) in Table 1 has been calculated from the various site mean intensities and susceptibilities for an inducing field of 0·45 oersteds. The values obtained are always greater than 1 for the Meldon sites and reach a maximum of 159 for the Type II intrusion samples from site 3. The samples from this site are also characterized by exceptionally high intensities and, as has been noted earlier, very low scatter.

It would appear from the results obtained so far that the Type II intrusion from site 3 has the most stable magnetization and should therefore represent the direction of the geomagnetic field existing at the time the rocks were magnetized. Comparison with the results of other workers (in Irving 1964) shows that there is no correlation with the

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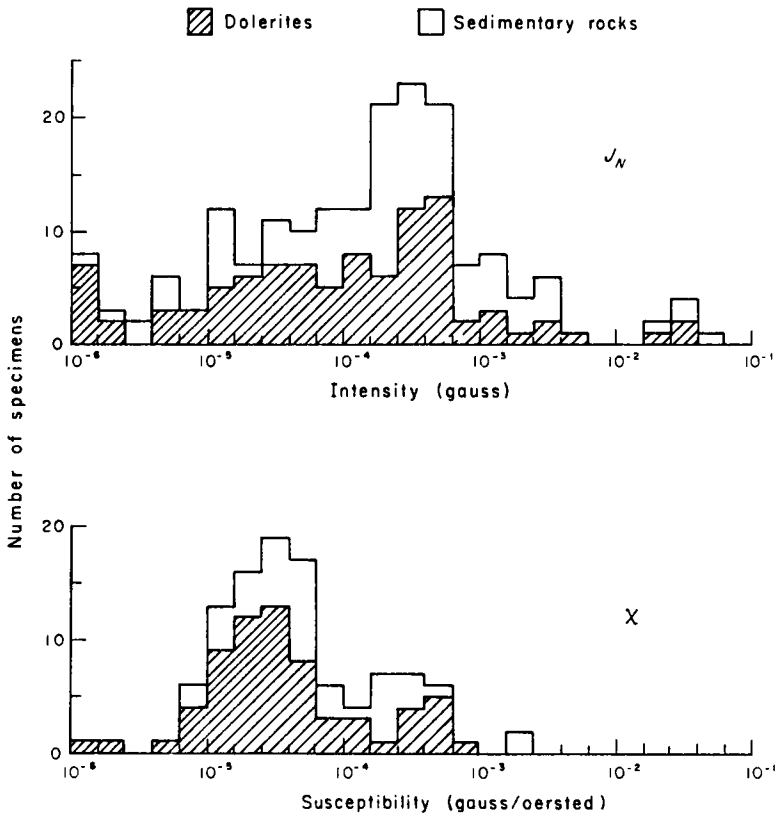


FIG. 4. Volume intensities of magnetization and susceptibilities of specimens from sites 1–5, Meldon. The abscissa scale is logarithmic.

direction of the early Permian field or with any of the various Carboniferous directions. The same appears to be true for the Group B directions. This point will be mentioned again later.

The five remaining sites in the Okehampton district all lie within one mile of the margin of the granite and the igneous and sedimentary rocks collected are similar to those in the Meldon Quarries. The high and variable intensities and susceptibilities recorded at three sites are also comparable with those recorded at Meldon but at two of these sites and at site 7, where the intensities are low, the directions of magnetization are too scattered for any site mean to be significant. The site mean of 159° + 12° for the Type II intrusion at site 8 can be compared with the Group A directions at the Meldon Quarries but is anomalous in that it is not similar to any of the established palaeomagnetic field directions. The mean of 210° − 25° at site 10 is possibly of greater significance as it is comparable with the mean direction of 189° − 21° calculated for four sites in the Tavistock district and believed to be an Upper Carboniferous field direction.

4.2. *Ferromagnetic mineral content*

The ferromagnetic mineral content of the samples was determined by an examination of a number of polished sections and from thermomagnetic balance curves. The polished sections show very clearly that by far the most important opaque mineral in both sedimentary and igneous rocks is pyrrhotite. It occurs mainly as large grains, several millimetres across, with very irregular, rounded boundaries. Minor amounts of chalcopyrite and magnetite are also present. X-ray photographs of the magnetic component extracted from crushed specimens showed typical pyrrhotite lines.

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Small fragments from fifteen samples of both igneous and sedimentary rocks from the Meldon Quarries were heated in a self-recording thermomagnetic balance. A maximum field of 2000 oersteds was provided by a small electromagnet. All the heating curves show well defined Curie temperatures between 300° C and 380° C. Those lying between 300° C and 320° C are the most common and the frequency then decreases until the 380° C Curie temperature was observed for one specimen only. A second decrease at *circa* 550° C was observed in eight cases while a Curie temperature of 630° C, indicating the presence of an impure haematite, was recorded for one sample. The well defined nature of the 300° C component and its occurrence in both sedimentary and igneous rocks suggest that it is due to pyrrhotite which is known to have Curie temperatures in the range 270–330° C (Haraldsen 1937). The *circa* 550° C component is most likely to be magnetite. Although pyrrhotite is commonly described as Fe_7S_8 it has a variable composition and the range of Curie temperatures is one result of this. Pyrrhotite is ferromagnetic in only part of the composition range (Nicholls 1955).

The cooling parts of the thermomagnetic curves are characterized by an increase in the contribution of the 550° C component with a corresponding decrease of the pyrrhotite component. The curves are therefore of a definitely irreversible type (Fig. 5). As the experiments were performed in air the most obvious explanation for this would be that the pyrrhotite has been oxidized and more magnetite formed.

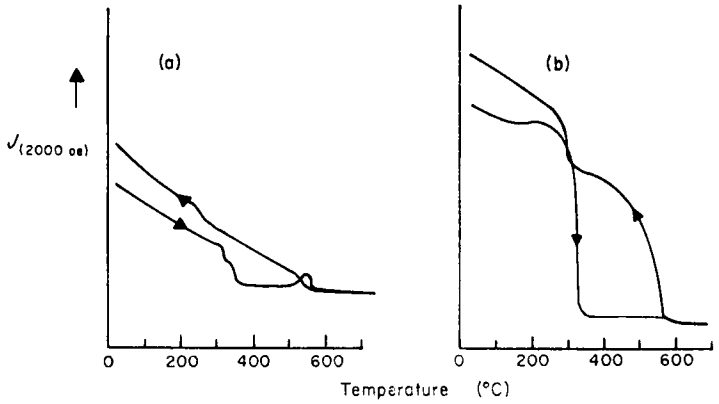


FIG. 5. Thermomagnetic curves for (a) Type II dolerite from site 3 and (b) tuff from site 4, Meldon.

Both pyrrhotite and magnetite Curie temperatures were discovered for samples from sites 8 and 9, where the high intensities are also suggestive of the presence of pyrrhotite. A weakly magnetized tuff from site 6 had only a single Curie temperature of 540° C.

4.3. *Thermal demagnetization*

The presence of pyrrhotite in samples from the Meldon Quarries and from two other sites in the Okehampton district has been established, and from the geological evidence it appears that this mineral must have been formed as a result of the metasomatism which was one of the later events in the intrusion of the granite. It is necessary, however, to show that pyrrhotite is significant as a carrier of remanent magnetization. To do this and also to test the stability of the magnetization, thermal and a.c. field demagnetization experiments were performed, the former using a vacuum furnace described elsewhere by the author (1965).

Some typical thermal decay curves are shown in Fig. 6. They have been chosen because they show the variations possible when samples covering the range of original intensities observed at Meldon are demagnetized at progressively higher temperatures.

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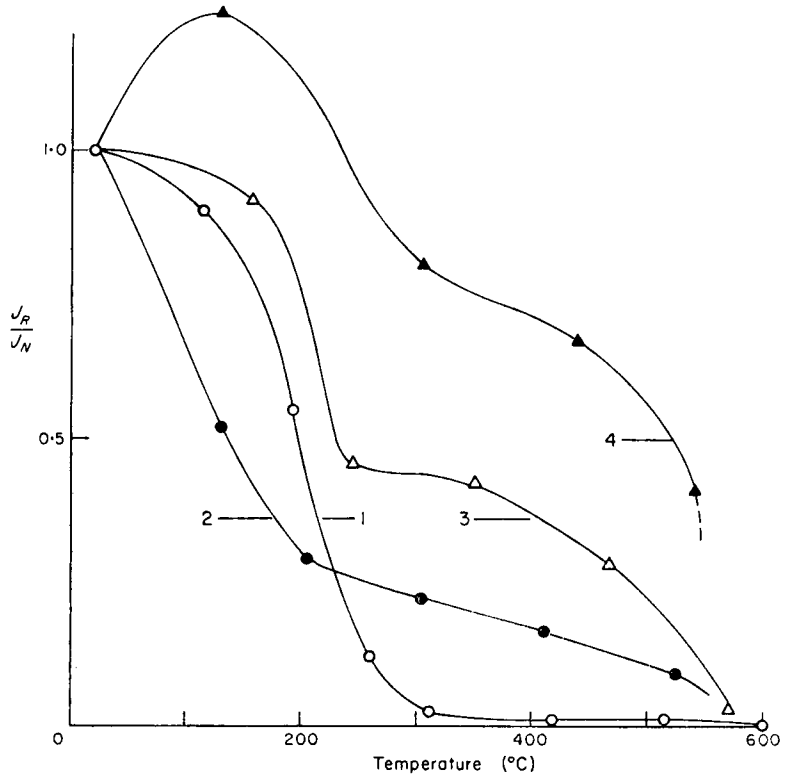


FIG. 6. Thermal demagnetization curves for (1) Type II intrusion (site 3), (2) black mudstone (site 2), (3) Type I intrusion (site 3) and (4) tuff (site 1).

Curve 1 in Fig. 6 was obtained from a strongly magnetized dolerite specimen ($3{\cdot}4 \times 10^{-2}$ gauss) and is characterized by a rapid decrease of the intensity up to 300° C followed by an interval in which no further change occurs, except perhaps for a small decrease near the magnetite Curie temperature. It is obvious from this that nearly all the remanence lies in the pyrrhotite and that once its Curie temperature is exceeded only a very small proportion of the original intensity remains (1% in curve 1). This higher temperature component could lie in the magnetite or it might simple represent the 'minimum intensity' of the specimen. Curves 2 and 3 (Fig. 6) were recorded for a sediment and an igneous rock of intermediate intensity ($5{\cdot}0 \times 10^{-4}$ gauss) and again show a range of blocking temperatures terminating at about 300° C, almost certainly due to pyrrhotite. The contribution of the second, higher temperature component is greater and in Curve 3 a Curie temperature of 580–600° C is indicated. The final result (curve 4) was obtained from a weakly magnetized specimen (1×10^{-5} gauss) of tuff and shows that most of remanence lies in the magnetite component.

The changes in the directions of magnetization are as complex as the decay curves, but a significant difference between the behaviour of the samples from the 860 S and 950 S bays has been found. The directions for the latter site are relatively stable and show little change until temperatures of 250° C are reached when the scatter suddenly increases (Fig. 8). The Group B directions of the 860 S samples are not so resistant to demagnetization and change rapidly in the same temperature interval. There is a tendency, however, for the latter to assume directions more comparable with those of the Group A type at about 250° C although the inclinations tend to be both upward and downward instead of the exclusively downward inclination of Group A. These results have been illustrated in the stereograms shown in Fig. 9. Thermal decay curves

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of a concave form (Fig. 6–2) tend to be more commonly associated with the rapid changes in direction but, as has been mentioned earlier, the shape of the curves is mainly dependent upon the original intensity. The directions most resistant to change by thermal demagnetization up to the Curie temperature of pyrrhotite are for the Type II intrusion of the 950 S bay and the corresponding decay curves are all similar to curve 1 in Fig. 6.

It can be concluded from these results that pyrrhotite is an important source of remanent magnetization in the Meldon rocks and that it is almost entirely responsible for the large values of the intensity of magnetization in some of the samples. Thermal demagnetization to temperatures in excess of 300° C completely destroys the remanence due to the pyrrhotite but not that due to the second component, which is most likely to be magnetite. In Fig. 9c the directions of magnetization measured after specimens had been heated to temperatures of *circa* 400° C are shown. The changes in the directions after demagnetization in the range 300–600° C tend to be large, and, as can be seen from Fig. 9c, no significant mean direction is apparent. Three specimens only were found to have directions which were resistant to treatment in this range but no conclusions can be reached from these.

4.4. *A.c. demagnetization*

The a.c. magnetic field demagnetization experiments were carried out on the apparatus described by Creer (1959), in which the steady decrease of current through the demagnetizing coil is achieved by using a variable electrolytic resistance.

The decay curves of the Meldon specimens are characterized by a rapid decrease of the intensity in low a.c. fields while progressively higher fields fail to produce any further changes (Fig. 7). The initial decrease is very marked and only 100 oersteds is sufficient to destroy at least 75% of the remanence. In normal magnetite-bearing rocks this feature is usually associated with unstable magnetization but it appears to

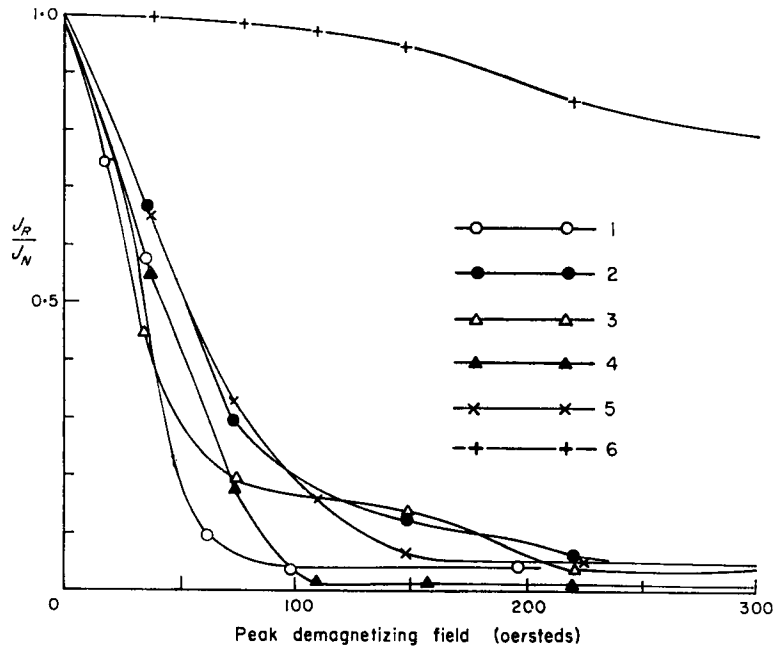


FIG. 7. A.c. demagnetization curves for (1) Type II intrusion (site 3), (2) Type I intrusion (site 2), (3) Type I intrusion (site 3) and (4) tuff (site 4). Curves 5 and 6 show the decrease of the TRM acquired by two specimens of the Type II intrusion (site 3) when cooling from 300° C and 600° C respectively in a field of 0·5 oersteds.

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be common in these pyrrhotite-rich rocks. The curve showing the decay of the artificial thermoremanent magnetization due mainly to the pyrrhotite component in a specimen of Type II dolerite is very similar to the NRM decay curves (Fig. 7). This TRM was produced by allowing the specimen to cool from 300° C in the Earth's field but the corresponding curve for another specimen from the same sample, after it had cooled from 600° C, shows that the latter TRM is much more resistant to a.c. demagnetization. The absence of any initial rapid decrease in the latter curve suggests that the pyrrhotite has been largely destroyed by the heating *in vacuo*.

The changes in the direction of magnetization associated with the initial drop in the intensity follow the pattern established by the thermal experiments. That is, the Group A directions are fairly stable while the Group B directions exhibit large changes but generally move to the opposite side of the stereogram (Fig. 9). The Type II intrusion from the 950 S bay was again found to be most resistant and the effects of a.c. demagnetization can be usefully compared with those of thermal demagnetization for this group. This has been done in Fig. 8 by plotting the precision of the mean direction

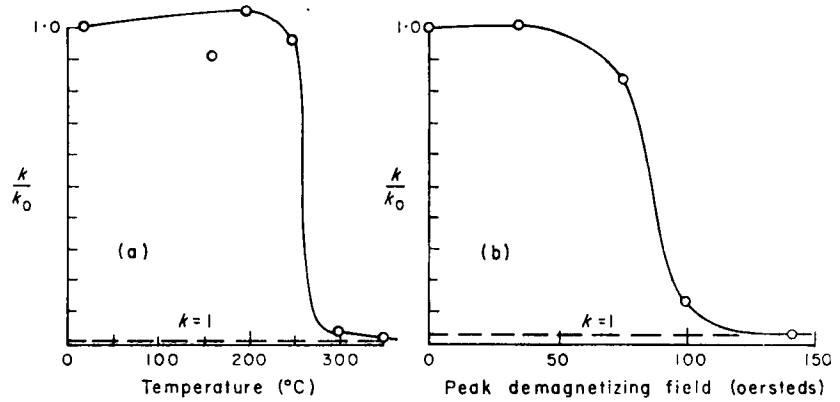


FIG. 8. The variation of the within-site scatter, indicated by the k value, with (a) thermal demagnetization and (b) a.c. demagnetization for four samples of the Type II intrusion (site 3). The scatter at the various stages of demagnetization (k) have been normalized against the scatter of the original directions of magnetization (k_0).

calculated from four sample means after demagnetization to various stages against the values of temperature and a.c. field respectively. The mean direction does not change appreciably by either method until the scatter rapidly increases. Fig. 8a shows clearly that pyrrhotite is responsible for the stable directions as the precision remains unchanged up to 250° C but as the Curie temperature of pyrrhotite (*circa* 300° C) is approached the directions become random. A field of only 125 oersteds is necessary to produce the same result; that is, by analogy, to completely destroy the remanence due to the pyrrhotite. A comparison of the corresponding decay curves shows that the pyrrhotite is relatively more resistant to thermal demagnetization than to a.c. demagnetization. Robertson (1963) has described demagnetization experiments on some pyrrhotite-bearing monzonite samples and finds that the directions become random after treatment in a field of 225 oersteds or heating to 340° C.

Demagnetization of the samples from sites 8 and 10 by thermal and a.c. magnetic field methods failed to improve the scatter of the directions of magnetization. The decay curves produced by both methods are similar to those recorded for samples from the Meldon Quarries and the remanence again appears to be due mainly to the pyrrhotite.

A.c. demagnetization in fields greater than 125 oersteds produces directions which change erratically and apparently randomly for all samples (Fig. 9c).

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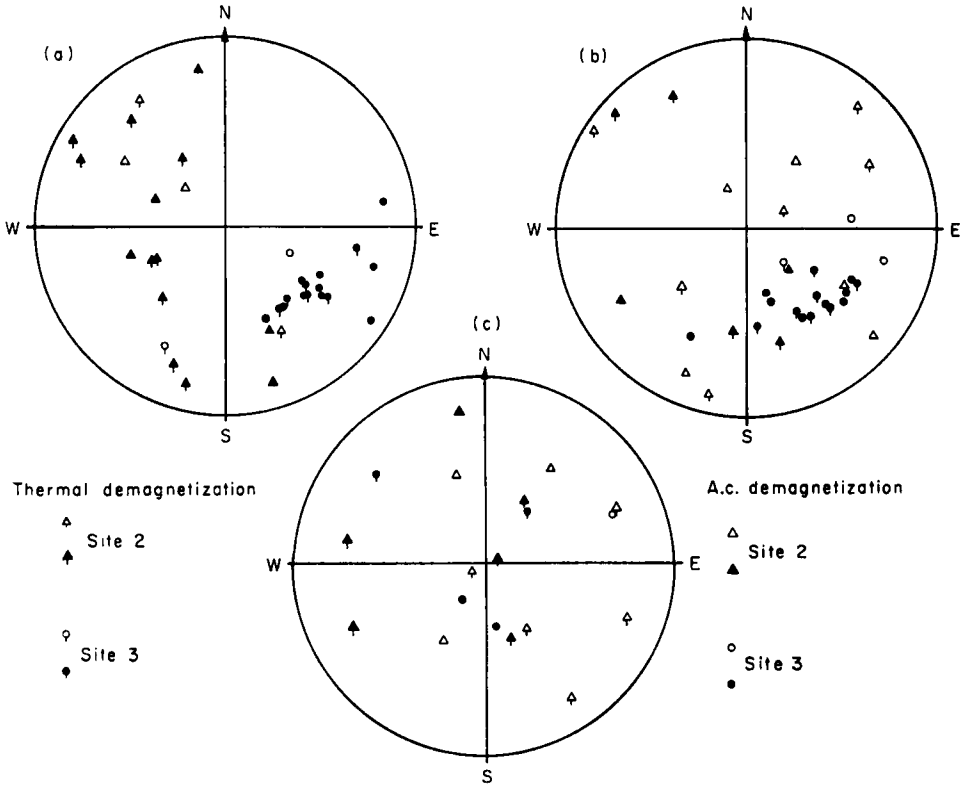


FIG. 9. Combined results of thermal and a.c. demagnetization on specimens from sites 2 and 3, Meldon. The original directions of magnetization (a) can be compared with the directions after demagnetization to 250° C or in a.c. fields of 50 oersteds (b) and (c) after demagnetization to 400° C or in fields of 150 oersteds.

4.5. *Anisotropy of susceptibility*

Pyrrhotite has been shown to be responsible for many of the magnetic properties of the igneous and sedimentary rocks at Meldon. In the relatively few cases in the literature where pyrrhotite has been proved as the source of NRM its properties have been anomalous, either in that it displays self-reversal (Almond *et al.* 1956, Everitt 1962, Robertson 1963, Bhimasankaram 1964) or anisotropy of susceptibility (Fuller 1963). No experimental evidence has been found for the Meldon samples which would suggest self-reversal although changes of about 180° were observed when demagnetizing samples from site 2. In any case an explanation is sought for groups of directions which differ by 90° and not 180°. Pyrrhotite, by virtue of its strong crystalline anisotropy, can become magnetized in a direction different from that of the applied field. Fuller (1963) has shown that the coincidence of the directions of magnetization with the cleavage of some slates from North Wales is due to the alignment of the high susceptibility basal planes of the pyrrhotite crystals with the cleavage plane. Only the component of the remanence lying in this plane has been preserved.

It seemed necessary, therefore, to investigate the possibility that the anomalous directions at Meldon could be due to an anisotropy of susceptibility. Measurements on twenty-nine specimens were made with a transformer bridge capable of detecting susceptibility differences of 1×10^{-6} gauss/oersted (Collinson *et al.* 1963). An anisotropy of the susceptibility with a mean maximum to minimum susceptibility ratio of 1·27 was detected in all but nine of these. Of these twenty specimens, fifteen have a plane of maximum anisotropy rather than well defined maximum and intermediate axes. The directions of the lower ends of the three axes of the anisotropy ellipsoi

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have been plotted in Fig. 10 for the samples in their present field positions. Comparing this figure with the directions of magnetization of the Meldon rocks (Fig. 3) it can be seen that there is an approximate correlation between the Group B directions and the directions of the maximum and, to a lesser extent, intermediate susceptibility axes. The latter appear in opposite quadrants of the stereogram because they have a small inclination and only the lower poles have been plotted. In contrast, the Group A directions coincide with the minimum susceptibility axes.

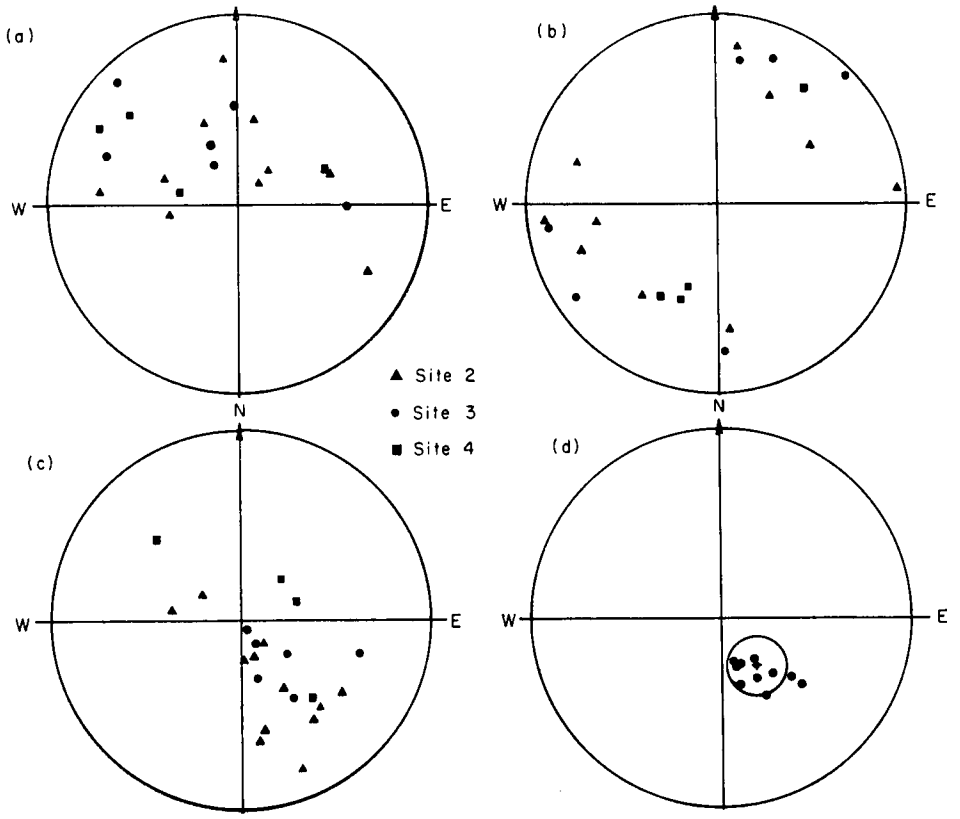


FIG. 10. Directions of the lower ends of (a) the maximum (b) the intermediate and (c) the minimum susceptibility axes of specimens from sites 2–4. In (d) the mean and circle of confidence of the minimum susceptibility axes of specimens from sites 2 and 3 are compared with the lower poles to the bedding planes.

The coincidence of the more unstable Group B directions with the directions of easy magnetization is in agreement with the general conclusions reached on the effect of anisotropy upon the remanent magnetization of rocks (Uyeda *et al.* 1963). Although the degree of anisotropy observed for the Meldon rocks would cause only small deflections of the remanence vectors according to the theoretical studies of Uyeda *et al.* (1963), it has been noted by Fuller (1963) that deflections observed in various rock types frequently exceed those predicted by the formula. It seems reasonable therefore to suggest that the Group B directions are the result of the deflection of the remanence vectors by the anisotropy. It is perhaps significant that the deflection appears to have been invariably towards the downward end of the maximum susceptibility axis. The directions found during the demagnetization experiments can then be interpreted as the result of the removal of the component of magnetization lying in the direction of

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easy magnetization and consequent emergence of the original component. Stacey (1960) has suggested that a.c. magnetic fields acting on rotated specimens can selectively cancel the deflection of the NRM due to the anisotropy but it appears from the experiments described here that thermal demagnetization is equally effective.

The anisotropy cannot fully explain the existence of the Group A and B directions of magnetization as it appears to be similar in size and orientation for samples from the 860 S and 950 S sites.

It can be seen from Fig. 10 that the directions of corresponding anisotropy axes are rather similar throughout the Meldon quarries. This is true for both igneous and sedimentary rocks and, as was the case with the directions of magnetization, correcting for the dip of the bedded rocks would result only in an increase in the scatter. This is not surprising as the secondary pyrrhotite is thought to be responsible for the anisotropy. In Fig. 10d the mean and circle of confidence of the minimum susceptibility directions of samples from the 950 S and 860 S bays are compared with the lower poles to the bedding planes of the sediments (or to the planes of the sheet-like intrusions), measured at the points where the samples were collected. The remarkably good agreement between the two suggests that the plane of maximum anisotropy is in fact the bedding plane. Dearman (1959), however, has shown that there is a well developed shear cleavage at Meldon which also coincides with the bedding. This cleavage was initiated at an early stage in the structural history and it seems likely that the stress system responsible for it also prevailed during the period of metasomatism and controlled to some extent the growth of the pyrrhotite grains. Pyrrhotite is generally formed with the basal planes of the pseudo-hexagonal crystals perpendicular to the direction of strain. As this plane is also the direction of easy magnetization the systematically orientated crystal growth would give rise to an anisotropy of susceptibility.

4.6. *Conclusions for the Okehampton district*

Pyrrhotite has been extensively developed in the rocks of the Okehampton district, particularly those at the Meldon Quarries, and its magnetization has largely obliterated the original NRM of both the sediments and the igneous rocks. According to the geological evidence the pyrrhotite was formed by the metasomatism occurring late in the history of the intrusion of the Dartmoor granite. The following features of the magnetization of the rocks can be attributed to the pyrrhotite:

(a) susceptibility and intensity values which vary considerably, sometimes over very short distances;

(b) remanent magnetization greater than the induced magnetization, sometimes a hundred times greater;

(c) similar directions of magnetization for both igneous and sedimentary rocks at a given site;

(d) anomalous and often scattered directions of magnetization;

(e) an anisotropy of susceptibility. The anisotropy ellipsoid is regularly orientated in the Meldon Quarries with the plane of maximum anisotropy coincident with the bedding planes or the planes of shear cleavage in the rocks;

(f) a.c. and thermal decay curves of rather distinctive shape.

The directions of magnetization cannot be explained fully at the present time. In the Meldon Quarries the directions can be divided broadly into two groups but the difference between them cannot be explained by the dip of the rocks or by any systematic difference in the magnetic properties of the rocks. The magnetization of the pyrrhotite must have been formed after the main intrusion of the Dartmoor granite when the rocks occupied nearly the same positions they do at the present but their directions of magnetization are not consistent with the direction of the palaeomagnetic field for that time (280 m.y.). The component remaining after heating to temperatures

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greater than 300° C might be regarded as the original magnetization of the rocks but it has random directions.

Although some of the directions at the Meldon Quarries have probably been affected by the anisotropy of susceptibility or even a self-reversal mechanism, the Group A directions, particularly the tightly grouped results from Type II intrusion, are relatively stable to demagnetization. The similarity in the ferromagnetic mineral content of the Group A and Group B rocks suggests that the direction difference might be connected with their history and not their composition. The only event post-dating the metasomatism was the breccia-faulting. Two large breccia-faults 200 ft apart and striking north 35° west have been located in the main Meldon Quarries (Dearman 1959). They have a north-easterly downthrow of at least 70 ft, the movement being mainly dip-slip. At the time the samples for this study were collected the north-east face of the 860 S bay lay south of these faults while the corresponding face of the 950 S bay was situated between them (Fig. 2). Dearman (1963) believes that faults of the type seen at Meldon were initiated during the Carboniferous and preferentially reactivated during mid-Tertiary times when the principal horizontal stress was directed north-north-east to south-south-west. One possible explanation of the Group A directions of the 950 S bay is that the rocks lying within the fault zones were remagnetized as a result of the relatively recent movements while those lying in the 860 S bay and displaying Group B directions were less affected. Similar breccia-faults exist east of Okehampton and the Sticklepath fault seen at the eastern margin of the map in Fig. 1 is still active today.

5. The remanent magnetization of samples from the Tavistock district

5.1. *The NRM and susceptibility*

The geology of the Tavistock district differs from that of the previously described Okehampton area to a small extent in its structural history and in the greater number of minor intrusions with sill forms. Five of the sites visited in the Tavistock district (Fig. 11) are located in these dolerite sills, one in the vesicular lavas lying near the top

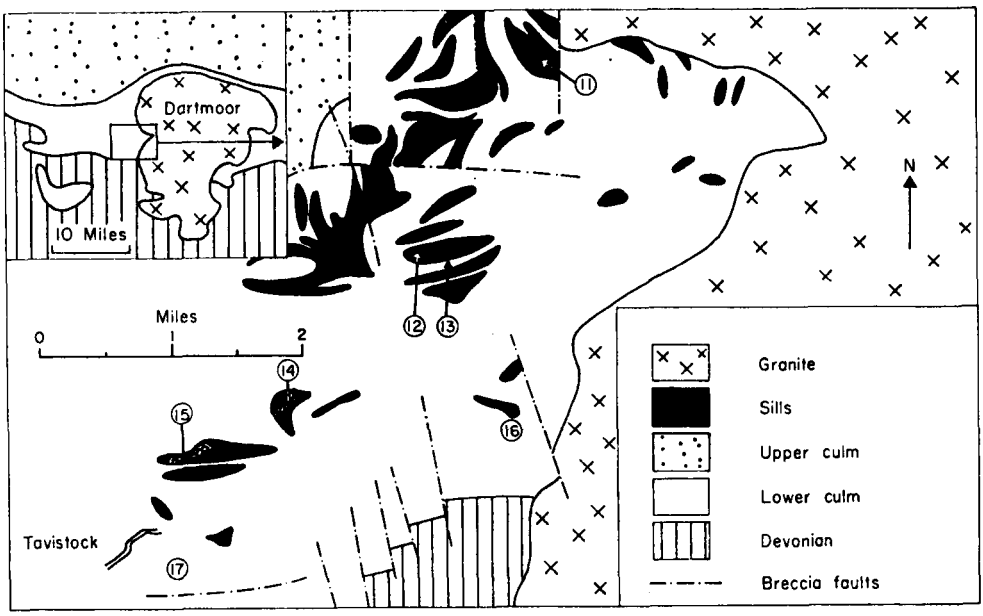


FIG. 11. Simplified geological map of the Tavistock district, Devonshire, showing the positions of collecting sites 11–17. (After Dearman & Butcher 1959.)

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of the Tavistock Calcareous Group and the seventh in the calc-flintas belonging to the same group. It can be seen from Table 1 that the sampling was particularly intensive at sites 14 and 15, where large quarries provide good exposures of the sills and the adjacent sediments.

In the Pitts Cleave Quarry (site 14) three lenticular sills with a combined thickness of at least 100 ft are interbedded with black shales. At the southern end of the 1000 ft long quarry face the sills and sediments are almost horizontal but the dips increase northwards until in the extreme north vertical sediments are encountered. It appears that the quarry lies in a single limb, at least 1000 ft long, of a recumbent fold with an east-west axis. The sampling was carried out at three points along this limb in order to examine the suggestion by Dearman & Butcher (1959) that the intrusion was indeed a sill and had been folded with the adjacent sediments. If this was true the directions of magnetization at the three points would be scattered unless the correction for the varying dip of the bedding planes was made. This should be true both for the sill itself and for the immediately adjacent sediments as the latter would have been heated by the intrusion and should therefore have acquired the same direction of magnetization. It was found that the lowest scatter occurred when all the directions were corrected to the present horizontal. Dip corrections for the three collecting points within the quarry differ by over 60° and if applied to the directions of magnetization would produce a site mean direction with a much larger circle of confidence than that given in Table 1. The similarity in the directions obtained from different parts of the sill is even more marked after the magnetic cleaning (Table 2). The most obvious explanation for this would be that the intrusion was formed after the sediments had been folded. It is suggested, however, that this is not the case but that the sill and the sediments were both remagnetized after they had been folded and the original NRM destroyed.

Only three-quarters of a mile away from this site another dolerite sill is exposed in the Wilminstone Quarry (site 15). The sill here is 200 ft thick and is tilted vertically in the hinge of a recumbent fold which plunges at 14° to the west-south-west. It has been intruded into black calcareous shales and the deformation structures seen in the dolerite and the sediment indicate that they were both involved in the folding.

The directions of magnetization of the samples from this quarry, particularly the sediments, are similar to those observed at the last site, when they are corrected to the present horizontal (Table 1). It is obvious that restoring the bedding to the horizontal at these two sites would only increase the between-site scatter.

The measurement of the dolerite samples from sites 14 and 15 have revealed intensity values which vary between $6{\cdot}7 \times 10^{-3}$ gauss and less than 1×10^{-6} gauss. Nine samples have mean intensities of less than 1×10^{-5} gauss and in some cases individual specimens had to be discarded because they were too weak to measure. There are corresponding but smaller variations in the susceptibilities. Large intensity variations have been noted earlier, in particular for samples from the Meldon Quarries, where they are explained as the result of the irregular distribution of secondary pyrrhotite in the rocks. The anomalously low intensity values of apparently unaffected dolerite samples will be mentioned later.

Three of the remaining sites are located on dolerite sills but they are natural exposures and the geology is therefore not known in such great detail as at Pitts Cleave and Wilminstone.

Site 13 is a small exposure of dolerite on the banks of the River Tavy. The sill lies within the thermal aureole of the Dartmoor granite, the edge of the granite lying a quarter of a mile to the east. The intensities of samples from this site are low but they are comparable with the minimum values recorded at sites 14 and 15. In spite of this the sample directions are grouped with relatively little scatter about a site mean of $172° - 16°$.

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Almost identical results were obtained from the dolerite samples from Smeardon Down (site 12). This sill also lies within the thermal aureole of the granite.

The second site on Smeardon Down is a small quarry which exposes laminated calc-flintas. Although only two samples were measured from this site they display remarkably different properties. One was found to have a negligible remanence while the second, collected only a few feet away and identical in appearance, had a mean intensity of $3{\cdot}5 \times 10^{-4}$ gauss.

Site 16 is a small dolerite tor lying 2000 ft from the granite margin (Fig. 11). The specimens measured have variable but generally high intensities of magnetization but the three mean sample directions are too scattered to give a reliable site mean. It seems likely that these effects are due to the presence of large quantities of pyrrhotite but the possibility that the outcrop, because of its exposed nature, has been the site of lightning strikes must also be considered.

The highly vesicular lavas at the final site were found to have a negligible remanent magnetization.

5.2. *Ferromagnetic mineral content*

In view of the widespread occurrence of pyrrhotite in the rocks of the Okehampton district and its importance there to the NRM attempts were made to distinguish this mineral in the Tavistock samples using the thermomagnetic balance. The variable intensities of magnetization for sites 14 and 15 (Table 1) give some indication that it might be present.

Curie temperatures of *circa* 300° C were recorded on the thermomagnetic curves for dolerite specimens from Wilminstone, Pitts Cleave and Cox Tor and for the strongly magnetized sediment from Smeardon Down and are taken as evidence for the existence of pyrrhotite. A Curie temperature of 520° C was also recorded for the Pitts Cleave specimen. Attempts to discover the ferromagnetic mineral content of other samples were not successful.

The presence of pyrrhotite was confirmed by an examination of polished sections of a strongly magnetized dolerite from Pitts Cleave and the Smeardon Down sediment.

5.3. *Thermal and a.c. demagnetization*

Specimens from sites 11–16 were demagnetized by thermal and a.c. magnetic field methods. The specimens from sites 14 and 15 showed a definite decrease in scatter after cleaning by both methods but there was no significant change in the mean site directions (Table 2). This result confirms the suggestion made earlier that there is no difference in the directions within the Pitts Cleave sills due to the variable dip or between the Pitts Cleave and Wilminstone site directions. The minimum scatter at both sites was produced after treatment in an a.c. field of 140 oersteds or after thermal demagnetization to 200° C, but the scatter again increased above 300° C. Owing to the low intensities of the NRM of many samples it was not possible to demagnetize all the specimens from the two sites. The cleaned mean directions given in Table 2 are based on the results from sixteen demagnetized specimens (eight samples) and the original directions of the remainder.

Table 2

Results of thermal and a.c. demagnetization on samples from sites 14 *and* 15. *One anomalous sample direction has been omitted from the calculation for site* 15

Site		$D°$	$I°$	$\alpha°$	k	N	
14	Pitts Cleave	193	−19	11	10	18	Original directions
		194	−19	10	14	18	After demagnetization
15	Wilminstone	204	−18	31	4	8	Original directions
		197	−26	13	19	8	After demagnetization

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Demagnetization of four specimens from sites 11 and 16 failed to produce any significant changes in the directions of magnetization. At the former site the difficulty of measuring the weak magnetization probably accounts for changes in the directions produced by heating in the vacuum furnace. The directions of the high intensity Cox Tor dolerite remained stable to thermal demagnetization until temperatures of over 250° C were reached and were also resistant to a.c. demagnetization. The latter result is surprising in view of the form of the a.c. decay curve (Fig. 13).

Some thermal and a.c. decay curves illustrating the variety of forms discovered are shown in Figs. 12 and 13. The former vary between the rapid intensity decrease of the Cox Tor dolerite and the more regular curve for the dolerite from Hill Bridge.

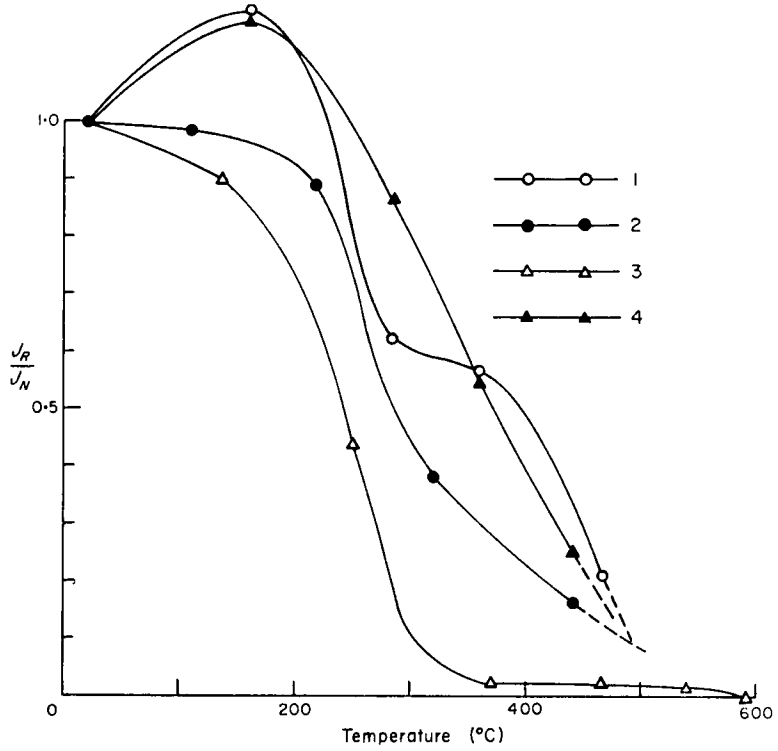


FIG. 12. Thermal demagnetization curves for (1) dolerite (site 14), (2) black mudstone (site 14), (3) dolerite (site 17) and (4) dolerite (site 11).

With the exception of the last result the curves indicate a definite range of blocking temperatures around 300° C, suggesting that pyrrhotite is responsible for part of the NRM. The contribution of a second, higher Curie temperature component (almost certainly magnetite) is also clearly indicated and the curves generally show similarities to those obtained for the Okehampton district rocks.

The forms of the a.c. decay curves are also variable but can be correlated with the original intensities of the specimens. The extremely rapid decrease of the initially high intensity of the Cox Tor dolerite is remarkable and is even more pronounced than in the curves obtained for the strongly magnetized Meldon specimens. This feature appears to be characteristic for samples containing much pyrrhotite, and corresponds to the thermal decay curves with a pronounced blocking temperature range terminating at 300° C. Decay curves 2 and 3 (Fig. 13) were obtained from two specimens cut from the same sample of dolerite but with quite different original moments. The specimen which gave curve 2 had an intensity of $4{\cdot}3 \times 10^{-2}$ gauss,

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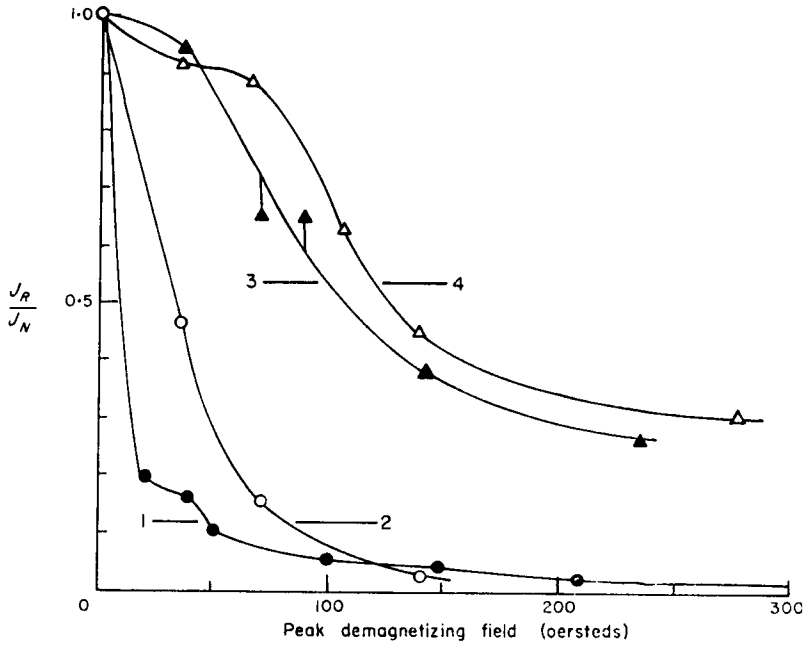


FIG. 13. A.c. demagnetization curves for dolerite specimens from (1) site 17, (2) site 14 and (3) site 14 and (4) site 11.

while the curve 3 specimen had an intensity of $3{\cdot}6 \times 10^{-4}$ gauss. This last example is similar to the decay curve (4) for a specimen of dolerite from site 13 which also had a relatively low intensity ($7{\cdot}5 \times 10^{-5}$ gauss). These lower intensity samples are obviously more resistant to demagnetization and this is explained as the result of a decreased pyrrhotite content and the consequent emergence of the second component.

These results obtained for the Tavistock samples are therefore interpreted as being due to a variable pyrrhotite content. It is suggested that all three samples from Cox Tor (site 16) and the higher intensity samples from sites 14 and 15 contain relatively large amounts of pyrrhotite while the dolerite at sites 12 and 13 are probably free of this mineral.

5.4. *Anisotropy of susceptibility*

The anisotropy of susceptibility of nine specimens from the Tavistock district were measured using a transformer bridge (Collinson *et al.* 1963). Six of these were found to be isotropic and the three remaining had maximum to minimum susceptibility ratios of 1·08, 1·12 and 1·11. It is possible to calculate limits to the mean anisotropy for all nine specimens by assuming a maximum difference of 1×10^{-6} gauss/oersted (the sensitivity limit of the transformer bridge) and a minimum difference of nil for the six apparently isotropic specimens. Calculated in this way the limits of the maximum to minimum susceptibility are 1·05 and 1·03 for the Tavistock specimens. A similar calculation for all the Okehampton district specimens gives the ratio limits of 1·21 and 1·18, suggesting that the degree of anisotropy is approximately related to the pyrrhotite content.

5.5. *Conclusions for the Tavistock district*

It has been shown in Table 2 that the directions of magnetization of both the sediments and the dolerite sills at Pitts Cleave (site 14) and Wilminstone (site 15) are identical. Similar directions have also been recorded for samples from sites 11, 12 and 13 (the first result was obtained from one sample only and will not be included in the following discussion). The rocks at these sites lie in very varying positions and

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it is obvious that correcting for the dip would considerably increase the between site scatter. There are two explanations for this:

(a) the sills were intruded after the sediments had been folded, or

(b) the sills were intruded before the folding but were subsequently remagnetized.

The presence of secondary pyrrhotite in some of the samples and the location of the sites within or near to the thermal aureole of the Dartmoor granite suggest that the second possibility is more likely. The geological evidence is also in favour of a pre-folding age of the intrusions. If this is true then the rocks should have been remagnetized at the time the granite was formed. This can be shown to be the case.

The Hercynian granites of south-west England have a mean age of 282 m.y. (Miller & Mohr 1964), the Dartmoor granite itself having been dated at 280 m.y. Although the granite is too weakly magnetized to give any palaeomagnetic directions the nearby Exeter lavas have been measured by Creer (1957) and Cornwell (1966). These lavas have been dated at 280 m.y., also by Miller & Mohr (1964). The mean direction of all the Exeter lava sites is $188°-13°$ with a circle of confidence of 7° and for the nine sites for which the dip correction can be made the mean is $189°-19°$ ($\alpha=10°$). The Tavistock site directions are therefore very similar, particularly to the latter result, the means for sites 12, 13, 14 and 15 being $189°-21°$ ($\alpha=13°$ and $k=49$).

6. The intensities and susceptibilities of the Tavistock and Okehampton samples

Large variations of the intensity of magnetization and the susceptibility of both igneous and sedimentary rocks from the Okehampton and, to a lesser extent, the Tavistock districts have been observed. These are explained as a result of the very variable pyrrhotite content of the rocks. The geological evidence suggests that the pyrrhotite was formed by the metasomatism which was one of the last events in the history of the intrusion of the Dartmoor granite. Only the fracturing which caused the breccia faults and the simultaneous hydrothermal activity are younger (Dearman 1959, Dearman & Butcher 1959). The emanations from the granite responsible for the metasomatic changes would be released mainly through fractures and other openings in the rocks. The changes would therefore be greatest in the vicinity of these channels but impregnations of pyrrhotite would also be formed in the intervening rock. It is thought that this is the explanation of the irregular distribution of the pyrrhotite.

It has also been shown that the lower limit for the intensities of the dolerite intrusions, particularly in the Tavistock district, is between 1×10^{-5} gauss and 1×10^{-6} gauss. Values of this order are exceptional for basic igneous rocks and might be regarded as another result of the metasomatic alteration of the rocks but for the fact that similar values have been recorded by Cornwell (1965) and Creer (1966) at ten other sites in south-west England. Nearly all of these low-intensity rocks are albite–dolerites, the most common type of basic intrusion in the Devonian and Carboniferous of Devonshire and Cornwall. The similarity of the mean direction of all the basic intrusions examined by Creer (1966) with the Permian palaeomagnetic field direction led him to suggest that the rocks had been remagnetized. However, the ultrabasic intrusions of Clicker Tor and Polyphant, Cornwall, have a mean direction ($185°+52°$) comparable with the Upper Devonian and Lower Carboniferous field directions and high intensities and susceptibilities (Cornwell 1965).

The reason for the low intensities must be simply the low magnetite contents of the albite–dolerites. The main opaque mineral in the samples examined is in fact ilmenite. The susceptibility of these low intensity samples is about 5×10^{-5} gauss/oersted and could probably be explained largely as the paramagnetic susceptibility of the iron-bearing silicates in the rocks.

Albite–dolerites and their volcanic equivalents, the spilites, are common in south-west England although they are petrologically unusual as they contain the albite or

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oligoclase feldspars typical for granites. Several explanations for the presence of these feldspars in basic rocks have been put forward, some of which suggest that the albite is secondary and was formed by a metasomatic process (Turner & Verhoogen 1960). Whether or not their low magnetite content is related to their petrological character is unknown but it is possible that it can be connected with their anomalous, Permian directions of magnetization.

7. Summary and conclusions

The igneous and sedimentary rocks of Lower Carboniferous age lying at the north-west border of the Dartmoor granite were extensively folded and altered during the Carboniferous. The magnetization of the rocks, particularly those from the Meldon Quarries, is due largely to the presence of secondary pyrrhotite and is characterized by a variable, and often large, remanence. In the Okehampton district the directions of magnetization are anomalous compared with known palaeomagnetic field directions and cannot be explained completely, although the anisotropy of susceptibility and a remagnetization due to fault movements may be responsible.

In the Tavistock district pyrrhotite is less important as a source of remanent magnetization and the directions of magnetization at the main sites are comparable with the Upper Carboniferous palaeomagnetic field direction. The rocks were remagnetized at that time, partly due to the introduction of the new ferromagnetic mineral, pyrrhotite, and partly to the increased temperature near the granite intrusion. The occurrence of metasomatic pyrrhotite implies temperatures of 400–600° C according to the evidence summarized by Deer *et al.* (1962), but there is still some argument about the validity of the geological thermometer used. There is no evidence that the Tavistock rocks have been generally heated to temperatures as high as 600° C but it is probable that some increase did occur during the metasomatism and may have remained long enough for the rocks to have acquired a moderate temperature viscous magnetization similar to that described by Chamalaun (1964). The lower pyrrhotite content and consequent low anisotropy of the Tavistock rocks and the absence of large breccia faults near the sites may be reasons why the directions of magnetization in this area are consistent with the palaeomagnetic field directions at the time of remagnetization.

Due to the mode of origin of the pyrrhotite the remanent magnetization of the Carboniferous rocks at the north-western border of the Dartmoor granite must be considered as an essentially local problem. Reviewing the results obtained here, however, and those described elsewhere in the literature, there appears to be a tendency for the remanent magnetization associated with pyrrhotite to be distinctive and often anomalous.

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